RIG-SMOOTH COMPLETE ALGEBRAS OVER A G-RING ARE COMPLETED FINITE TYPE ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete, from the free resolution of the presentation to the final isomorphism via the close-enough threshold

Editorial note. The label proposition-approximate names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- presentations of complete algebras and free resolutions of the quotient
- the henselization of a Noetherian pair and its G-ring stability under polynomial rings
- Artin approximation for the henselization to completion map of a G-ring pair
- the close-enough lemma giving a congruence threshold under which quotient completions are isomorphic
- the easy algebraization lemma identifying a completion of a quotient of the henselization with a completed finite type A-algebra

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 1477; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAS: sources/stacks-restricted/pinned-original-source.tex, source lines 1452–1452. Prerequisite pointer invoked at proof line(s) 1528; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAR: sources/stacks-restricted/pinned-original-source.tex, source lines 1363–1363. Prerequisite pointer invoked at proof line(s) 1515; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A02: sources/stacks-chow/tags.txt, source lines 12620–12620. Prerequisite pointer invoked at proof line(s) 1503; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGV: sources/stacks-chow/tags.txt, source lines 13208–13208. Prerequisite pointer invoked at proof line(s) 1507; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 07PV: sources/stacks-chow/tags.txt, source lines 9810–9810. Prerequisite pointer invoked at proof line(s) 1510; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AH5: sources/stacks-chow/tags.txt, source lines 13218–13218. Prerequisite pointer invoked at proof line(s) 1514; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let I be an ideal of a Noetherian G-ring A. Let B be an object of (0AL4). If B is rig-smooth over (A, I), then there exists a finite type A-algebra C and an isomorphism $B \cong C^\wedge$ of A-algebras.*

Proof. Choose a presentation $B = A[x_1, \ldots, x_r]^\wedge/J$. Write $P = A[x_1, \ldots, x_r]^\wedge$. Choose generators $g_1, \ldots, g_m \in J$. Choose generators $k_1, \ldots, k_t$ of the module of relations between $g_1, \ldots, g_m$, i.e., such that

$$P^{\oplus t} \xrightarrow{k_1,\ldots,k_t} P^{\oplus m} \xrightarrow{g_1,\ldots,g_m} P \to B \to 0$$

is a resolution. Write $k_i = (k_{i1}, \ldots, k_{im})$ so that we have

$$\sum\nolimits_j k_{ij} g_j = 0 \tag{0.1.1}$$

for $i = 1, \ldots, t$. Denote $K = (k_{ij})$ the $m \times t$-matrix with entries k_{ij}.

Let $A[x_1, \ldots, x_r]^h$ be the henselization of the pair $(A[x_1, \ldots, x_r], IA[x_1, \ldots, x_r])$, see More on Algebra, Lemma 0A02. We may and do think of $A[x_1, \ldots, x_r]^h$ as a subring of $P = A[x_1, \ldots, x_r]^\wedge$, see More on Algebra, Lemma 0AGV. Since A is a Noetherian G-ring, so is $A[x_1, \ldots, x_r]$, see More on Algebra, Proposition 07PV. Hence we have approximation for the map $A[x_1, \ldots, x_r]^h \to A[x_1, \ldots, x_r]^\wedge = P$ with respect to the ideal generated by I, see Smoothing Ring Maps, Lemma 0AH5. Choose a large enough integer n as in Lemma 0GAR. By the approximation property we may choose $g'_1, \ldots, g'_m \in A[x_1, \ldots, x_r]^h$ and a matrix $K' = (k'_{ij}) \in \mathrm{Mat}(m \times t, A[x_1, \ldots, x_r]^h)$ such that $\sum_j k'_{ij} g'_j = 0$ in $A[x_1, \ldots, x_r]^h$ and such that $g_i - g'_i \in I^n P$ and $K - K' \in I^n \mathrm{Mat}(m \times t, P)$. By our choice of n we conclude that there is an isomorphism

$$B \to P/(g'_1, \ldots, g'_m) = \big(A[x_1, \ldots, x_r]^h/(g'_1, \ldots, g'_m)\big)^\wedge$$

This finishes the proof by Lemma 0GAS. □